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Detailed mark scheme

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AQA A Level Further Mathematics (7367)

Topic D: Further Algebra and Functions

Sub topic D4: Method of Differences - Use of Partial Fractions

Mark Scheme

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AQA A Level Further Mathematics (7367)

Mark Scheme

Topic	**D: Further Algebra and Functions**
Sub topic	**D4: Method of Differences - Use of Partial Fractions**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms equation using the numerators from each side	AO1.1a	M1	$\frac{2}{(r+1)(r+2)(r+3)} \equiv \frac{A}{(r+1)(r+2)} + \frac{B}{(r+2)(r+3)}$
	Obtains the correct values of A and B	AO1.1b	A1	$\Rightarrow 2 \equiv A(r+3) + B(r+1)$ $\Rightarrow A = 1, B = -1$
(b)	Uses 'their' result from part (a) to write fraction as sum of differences Ignore $\frac{1}{2}$ at this stage.	AO1.1a	M1	$\sum_{r=9}^{97} \frac{1}{(r+1)(r+2)(r+3)}$ $= \frac{1}{2}\sum_{r=9}^{97} \frac{1}{(r+1)(r+2)} - \frac{1}{(r+2)(r+3)}$
	Clearly shows step of cancelling of terms	AO2.4	M1	$\sum_{r=9}^{97} \frac{1}{(r+1)(r+2)} - \frac{1}{(r+2)(r+3)}$
	Obtains correct two term difference Ft 'their' values for A and B provided that 'their' A = −'their' B	AO1.1b	A1	$= \frac{1}{10\times 11}$ $- \frac{1}{11\times 12} + \frac{1}{11\times 12}$
	States that method of differences gives $2\times\sum_{r=9}^{97} \frac{1}{(r+1)(r+2)(r+3)}$ so divides their answer by 2 to obtain correct rational solution from fully correct working **AG** (If student merely divides by 2 without justification withhold this mark)	AO2.1	R1	$- \frac{1}{12\times 13} + \frac{1}{12\times 13}$ … $- \frac{1}{98\times 99} + \frac{1}{98\times 99}$ $- \frac{1}{99\times 100}$ $= \frac{1}{10\times 11} - \frac{1}{99\times 100}$ $= \frac{89}{9900}$ $\therefore \sum_{r=9}^{97} \frac{1}{(r+1)(r+2)(r+3)}$

			$= \frac{1}{2} \times \frac{89}{9900}$ $= \frac{89}{19800}$
			Total 6 marks

Q2.

(a) $\frac{1}{r(r+2)} = \frac{A}{r} + \frac{B}{r+2}$

M1

$A = \frac{1}{2}, \quad B = -\frac{1}{2}$

ft incorrect A

A1,A1F

3

(b) $r = 1 \quad \frac{1}{1.3} = \frac{1}{2}\left(\frac{1}{1} - \frac{1}{3}\right)$

$r = 2 \quad \frac{1}{2.4} = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{4}\right)$

$r = 3 \quad \frac{1}{3.5} = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{5}\right)$

3 rows (PI) numerical values only

M1

$r = 4 \quad \frac{1}{48.50} = \frac{1}{2}\left(\frac{1}{48} - \frac{1}{50}\right)$

Last row – could be implied

A1F

Cancelling appropriate pairs

M1

$\text{Sum} = \frac{1}{2}\left(\frac{1}{1} + \frac{1}{2} - \frac{1}{49} - \frac{1}{50}\right)$

Allow if the $\frac{1}{2}$ is missing only

A1F

$= \dfrac{894}{1225}$

CAO (or equivalent fraction)

A1

5

[8]

Q3.

(a) $A = \dfrac{1}{2}, B = -\dfrac{1}{2}$

For either A or B

B1,

For the other

B1F

2

(b) Method of differences clearly shown

M1

$\text{Sum} = \dfrac{1}{2}\left(1 - \dfrac{1}{2n+1}\right)$

A1

$= \dfrac{n}{2n+1}$

AG

A1

3

(c) $\dfrac{1}{2(2n+1)} < 0.001$ or $\dfrac{n}{2n+1} > 0.499$

Condone use of equals sign

M1

$1 < 0.004n + 0.002$ or $n > 0.998n + 0.499$

$n > \dfrac{0.998}{0.004}$ or $0.004n > 0.998$

OE

A1

$n = 250$

ft if say 0.4999 used
If method of trial and improvement used, award full marks for a completely correct

solution showing working

A1F

3

[8]

Q4.

(a) $1 = A(r + 2) + Br$

M1

$2A = 1, \quad A = \frac{1}{2}$

A1

$A + B = 0, \quad B = -\frac{1}{2}$

A1

3

(b) $r = 10 \quad \frac{1}{2}\left(\frac{1}{10.11} - \frac{1}{11.12}\right)$

if (a) is incorrect but $A = \frac{1}{2}$ *and* $B = -\frac{1}{2}$ *used, allow full marks for (b)*

$r = 11 \quad \frac{1}{2}\left(\frac{1}{11.12} - \frac{1}{12.13}\right)$

....

$r = 98 \quad \frac{1}{2}\left(\frac{1}{98.99} - \frac{1}{99.100}\right)$

3 relevant rows seen

M1A1

$S = \frac{1}{2}\left(\frac{1}{10.11} - \frac{1}{99.100}\right)$

if split into $\frac{1}{2r} - \frac{1}{r+1} + \frac{1}{2(r+2)}$ *follow mark scheme, in which case*

$\frac{1}{2.10} - \frac{1}{2.11} + \frac{1}{2.100} - \frac{1}{2.99}$ *scores m1*

m1

$= \frac{89}{19800}$

A1

4

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Q5.

(a) $r^2 + r - 1 = A(r^2 + r) + B$

Any correct method

M1

$A = 1$, $B = -1$

A1

ft B if incorrect A and vice versa

Or $\frac{r^2+r-1}{r^2+r} = 1 - \frac{1}{r(r+1)}$ *B1*

$= 1 - \left(\frac{1}{r} - \frac{1}{r+1}\right)$ *M1A1*

A1F

3

(b)

$r = 1 \quad 1 - \frac{1}{1} + \cancel{\frac{1}{2}}$

$r = 2 \quad 1 - \cancel{\frac{1}{2}} + \cancel{\frac{1}{3}}$

$r = 99 \quad 1 - \cancel{\frac{1}{99}} + \frac{1}{100}$

Do not allow M1 if merely $\sum \frac{1}{r} - \sum \frac{1}{r+1}$ *is summed*

M1

A1 for suitable (3 at least) number of rows

A1F

Sum = 98 + $\frac{1}{100}$

Must have 98 or 99

m1

= 98.01

OE Allow correct answer with no working 4 marks

A1F

4

[7]

Examiner reports

Q2.

This question was generally well done, with many candidates scoring full marks. When errors did occur they were usually in the omission of one of the four fractions that made up the sum, notably $\frac{1}{98}$.

Q3.

Almost all candidates produced correct solutions to parts (a) and (b), apart from the odd arithmetical slip.

There was, however, less success with part (c). Few candidates worked with inequalities (although the use of the equals sign was condoned) and the lack of ability to solve an equation in n with decimals involved led to the solutions for n which common sense should have told candidates was impossible. It was not infrequent to see n as a decimal less than unity and, even when candidates, using equalities, arrived at 249.5, they left it as their final answer, not considering that n had to be integral.

Q4.

A common error in part (a), when finding the values of A and B, was to write $1 = A(r + 1)(r + 2) + B(r + 1)r$. This, in turn, frequently led to a correct answer for A (by incorrect means) but an incorrect value for B. In this case candidates tried to cancel out terms when in fact because of their incorrect value for B, their terms did not cancel out and so were unable to gain credit. A few candidates realised that their value for B was incorrect and that it had to be $-\frac{1}{2}$ and so proceeded to do part (b) correctly. Generally speaking, those candidates who found A and B correctly went on to score full marks. If they didn't, it was usually due to an incorrect value of r being substituted either at the beginning or the end of the series.

Q5.

Many candidates experienced difficulty in finding the values of A and B in part (a). They seemed to want C D to equate the left hand side of the identity to $\frac{C}{r} + \frac{D}{r+1}$, thus ignoring the fact that the powers of r in the numerator and denominator were equal. Generally the most successful candidates were those who rewrote the left hand side of the equation as $1 - + \frac{1}{r(r+1)}$ with the subsequent expressing of $\frac{1}{r(r+1)}$ in r(r + l) r(r + l) partial fractions.

If candidates were successful in finding the values of A and B, they usually went on to complete part (b) correctly. The main source of error in this part, if mistakes were made, was to overlook the fact that that the constant term, as well as the variable terms, had to be summed from 1 to 99. The constant term was often left as 1.